

"The most beautiful X-ray photographs ... ever taken."

J. D. Bernal, 1958

Photograph 51 produced the first clear image of the crystalline structure of DNA, and has been heralded as one of science's most important photographs.

construct their own 3D model of DNA. Meanwhile, at King's College, London, Franklin was recruited to take over DNA research using X-ray techniques. This caused tension with the former leader of the research team, Maurice Wilkins.

After refining her X-ray techniques, in 1952 Franklin produced the first clear image of DNA, labeled Photograph 51. From the "X" shape that her image revealed, she noted that the structure of DNA was a helix (spiral) and that it was formed of two strands, not three as previously believed. Franklin also proposed the "backbone" of DNA lay on the outside of the molecule.

In early 1953, and without Franklin's knowledge, Maurice Wilkins showed Photograph 51 to Crick and Watson, as well as Franklin's unpublished thoughts about the shape of DNA. These helped the pair confirm DNA's double helix structure—they completed their model and published the results in April 1953, but Franklin's key contributions went unrecognized.

Beyond DNA

Franklin moved to Birkbeck College, London, in 1953 to study the crystalline structure of viruses. Applying her skill as a crystallographer, she proposed a groundbreaking new model for the molecular structure of the tobacco mosaic virus (TMV). This was proven to be correct,

